

SOFTWARE DEVELOPER

Software developer with over five years of experience in both large and small-scale companies specializing in data visualization as well as full-stack development of complex single-page and multi-page web apps, with experience in mobile app development.

TECHNICAL PROFILE

PROGRAMMING LANGUAGES & FRAMEWORKS

JavaScript, TypeScript, Vue, Nuxt, VitePress, Angular, React, React Native, and Node.js | C#, ASP.NET, .NET Core, and Xamarin | CSS, Tailwind CSS, PostCSS, and SCSS | HTML and XML | SQL, PostgreSQL, MySQL, and SQLite | Java | Kotlin | MatLab | Python and Flask | C and C++ | R and R Studio

DEVOPS, TOOLS & PLATFORMS

AWS, S3, DynamoDB, CloudFront, and Lambda | Azure DevOps, WebApps, Pipelines, Repos, Boards, Monitor, Blob Storage, Data Lake, and Functions | CircleCI | Git and GitHub | Mapbox | QGIS | Figma

RELATED PROFESSIONAL EXPERIENCE

Software Developer | Darkhorse Analytics | Edmonton, AB

May 2022 – Feb 2026

Developed data visualization tools for researchers, nonprofits, and corporate clients to bring their data to life. Primarily contributed as a software developer but also participated in design and data analytics.

- Created static and animated charts, interactive reports, and dashboards to visualize data over time and other variables using as many as 9 million data points
- Used Vue and Nuxt to develop and maintain performant data visualization tools, including interactive maps spanning as many as 172 countries with as many as 70,000 regions per country.
- Developed and maintained 4 internal npm libraries for commonly used features: Tutorials, Tooltips, Search, and Image Export
- Created, maintained, and ran data pipelines for multiple projects, and helped to create and expand multiple custom CMS tools
- Worked with the design team using Figma to create and modify designs for custom UI components and layouts
- Communicated with clients to help them identify project requirements and to provide progress updates over time, receiving and addressing feedback throughout the project life cycle

Software Developer Co-op Student | PCL | Edmonton, AB

May 2020 – Apr 2022

Technical Solutions Analyst Co-op Student | PCL | Edmonton, AB

Jan 2019 – Sep 2019

Worked individually as the sole developer and designer on projects focused on research and development. Later worked as a member of a larger team on new products to be integrated into a preexisting product ecosystem.

- Used early versions of Autodesk Forge APIs (now called Autodesk Platform Services) to develop a web application to view 3D models, track version history, and render embedded parameterized models in external applications
- Developed web applications using Angular with .NET Core servers
- Developed mobile applications across iOS and Android using Xamarin
- Created prototype hardware to mimic a relay-controlled gate and developed custom APIs to control it
- Developed a Revit Add-In to facilitate data extraction from large models

FORMAL EDUCATION

Bachelor of Science in Computer Engineering | University of Alberta | Edmonton, AB

2022

Completed a comprehensive, five-year Computer Engineering Co-op Program with a focus on software development, including 5 private sector work terms lasting 4 months each.

INVOLVEMENT:

- Participated in the UofA Programming Contest Club and the Annual HackED Hackathon from 2018 to 2022

SELECTED PROJECT EXPERIENCE

Global Opportunity Atlas

2025

<https://global.opportunityatlas.org/>

Combined data from five different research teams to visualize how a child's opportunity for upward mobility differs depending on where they were born. Addressed significant technical hurdles by developing an interactive mapping interface capable of dynamically rendering data for indicators and demographic filters that differed based on regional data collection methods. Implemented flexible frontend and backend systems to account for these differences, and to allow easy configuration through a custom CMS.

AIDSVu & HepVu

2024

<https://map.aidsvu.org/> & <https://map.hepvu.org/>

Contributed to the development and maintenance of web applications for AIDSVu and HepVu, which are public health platforms used for national disease surveillance. Played a key role in unifying both projects into a single Nuxt/Vue.js codebase, implementing a comprehensive configuration system to handle distinct branding, content, and data requirements that differed significantly between the two websites. Specifically addressed this challenge for reports by developing a custom YAML configuration system. This decoupled report content and formatting from the core code, allowing for dynamic rendering and streamlined data updates across both apps.

Lumina Foundation – A Stronger Nation

2024

<https://strongernation.luminafoundation.org/>

Contributed to the development of a suite of eight interactive charts and visualizations to track national progress toward the goal of increasing credentials of value attainment to 75% of adults in the U.S. labour force by 2040. A key contribution involved solving significant responsive design challenges, ensuring that data visualizations and interactive controls remained performant and intuitive across both mobile and desktop environments.

The Opportunity Atlas - Mobility Trends Update

2024

<https://www.opportunityatlas.org/>

Collaborated on a large-scale application expansion to integrate "Module 2", introducing new mobility trend data and a new "Trend and Gap Analysis" feature. Contributed to the strategic migration of legacy React components into a modern Vue.js architecture and refactored technical debt. Bridged disparate tech stacks to improve the platform's long-term maintainability and performance, ensuring both new and legacy modules could coexist within a unified, scalable codebase.

The U.S. Climate Vulnerability Index

2023

<https://map.climatevulnerabilityindex.org/>

Collaborated on the development of a massive geospatial data platform aggregating 180+ datasets to visualize climate vulnerability across 70,000 U.S. census tracts and counties. Implemented a novel tree-based navigation system for complex indicator selection and engineered a dynamic, animated tutorial framework. This tutorial system later served as the architectural foundation for a new internal library, subsequently used within other Darkhorse projects including AIDSVu, HepVu, and the Global Opportunity Atlas.